

# INTEGRAL CURVATURE ESTIMATES AND GEODESIC LIMITS FOR STABLE ABELIAN HIGGS FIELDS

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**ABSTRACT.** We prove that, on closed three-manifolds, the limit interface of a sequence of stable abelian Higgs critical points with uniformly bounded energy is a finite union of smooth closed immersed geodesics, possibly with multiplicity and self-intersections. This follows from an integral estimate for the diffuse curvature, derived from an integral bound on the discrepancy.

**AI usage statement.** Inspired by the recent work of the first author on the classification of entire stable solutions, we felt it should be possible to classify limits of stable solutions in closed manifolds using geometric measure theory, without developing a strong regularity theory. We prompted the LLM Astra-6 to explore the possibilities in that direction. The LLM produced a first proof with little help from us. We checked and rewrote it in order to highlight its novel mathematical ideas and improve the clarity of the presentation.

## 1. INTRODUCTION

Diffuse approximations connect nonlinear elliptic equations with the geometry of minimal submanifolds. In codimension one, this connection goes back to the work of Modica and Mortola [11] on the Allen–Cahn equation. In codimension two, the abelian Higgs model plays an analogous role. Let  $(M, g)$  be a smooth oriented Riemannian manifold, let  $\Omega \subset M$  be open, and let  $L \rightarrow \Omega$  be a Hermitian line bundle. We consider the self-dual Yang–Mills–Higgs energy

$$\begin{aligned} E_\varepsilon(u, \nabla; \Omega) &:= \frac{1}{2} \int_{\Omega} e_\varepsilon(u, \nabla) \, d\text{vol}_g, \\ e_\varepsilon(u, \nabla) &:= |\nabla u|^2 + \varepsilon^2 |F_\nabla|^2 + \frac{1}{4\varepsilon^2} (1 - |u|^2)^2, \end{aligned} \tag{1.1}$$

where  $u$  is a section and  $\nabla$  is a smooth unitary connection with real curvature  $F_\nabla = i\nabla^2$ . A stable critical point in  $\Omega$  is defined as a pair  $(u, \nabla)$  satisfying

$$\delta E_\varepsilon(u, \nabla) = 0, \quad \delta^2 E_\varepsilon(u, \nabla) \geq 0,$$

where both variations are tested against fields compactly supported in  $\Omega$ . The compactness theorem of Pigati and Stern [16] shows that, under a uniform energy bound, the energy measures of sequences of critical points with  $\varepsilon_k \rightarrow 0$  converge subsequentially to the weight measure of a stationary integral varifold of codimension two, up to a fixed normalisation. This result paved the way to many recent advances in the existence of solutions, regularity and geometric evolution [3, 4, 14, 9, 8, 12].

Stability has proved especially effective in codimension-one regularity theory [18, 19, 22, 20, 21, 10, 7], but its geometric consequences in codimension two remain less understood. Here we establish a regularity theorem for stable abelian Higgs fields in dimension three.

**Theorem A.** *Let  $\Omega$  be an open subset of a smooth oriented Riemannian three-manifold  $(M^3, g)$ . Consider sequences  $\varepsilon_k \rightarrow 0$  and  $(u_k, \nabla_k)$  of smooth stable critical points of (1.1) with  $\varepsilon = \varepsilon_k$ ,  $|u_k| \leq 1$  and*

$$\sup_k E_{\varepsilon_k}(u_k, \nabla_k; \Omega) \leq \Lambda.$$

*Then, for every  $\Omega' \Subset \Omega$ ,*

*(1) The discrepancies*

$$\xi_k := \varepsilon_k |F_{\nabla_k}| - \frac{1 - |u_k|^2}{2\varepsilon_k}$$

*satisfy  $\|\varepsilon_k^{-1} \xi_k\|_{L^1(\Omega')} \rightarrow 0$ .*

*(2) Define  $\nu_k := (*F_{\nabla_k})^\sharp / |F_{\nabla_k}|$  where  $F_{\nabla_k} \neq 0$ , and choose any measurable unit extension to its zero set. For the stress-energy tensor defined in (2.8),*

$$\lim_{k \rightarrow \infty} \|T_{\varepsilon_k} - e_{\varepsilon_k} \nu_k^\flat \otimes \nu_k^\flat\|_{L^1(\Omega')} = 0. \quad (1.2)$$

Moreover, the divergence-free vector field  $q_{\varepsilon_k}$  associated with the modified Jacobian in (4.1) satisfies

$$\limsup_{k \rightarrow \infty} \int_{B_r(x)} |\nabla_{q_{\varepsilon_k}} \nu_k| d\text{vol}_g \leq C\sqrt{r} \quad (1.3)$$

for every normal ball  $B_{2r}(x) \subset \Omega'$  with  $0 < 2r < r_0$ . The integrand is extended by zero where  $q_{\varepsilon_k} = 0$ . The constants depend on  $\Lambda$ , the fixed interior domains and their geometry.

- (3) Every subsequential limit varifold supplied by [16] is locally a finite union of smooth geodesic arcs without interior endpoints, counted with positive integer multiplicities and possibly intersecting.

If  $\Omega = M$  is closed, each such limit is a finite union of closed immersed geodesics, counted with positive integer multiplicities.

Theorem A, which follows from Theorem 3.1, Propositions 4.1 and 4.2 and Theorem 4.3 below, concerns the geometry of the limiting varifold at arbitrary multiplicity. It does not require a multiplicity-one hypothesis or exclude intersections of distinct geodesics. Its proof has two main parts.

The analytic starting point is the the first author's work on entire stable abelian Higgs fields [2]. In that setting, appropriate energy growth and expanding logarithmic cutoffs make the errors in a stability comparison vanish, leading to exact identities and classification. We use the same underlying test pairs with fixed compactly supported cutoffs. The cutoff errors remain, and ambient curvature introduces additional terms, so the comparison must be used quantitatively.

The discrepancy  $\xi_\varepsilon$  compares the curvature and potential amplitudes. The localized stability calculation, together with the critical-point estimates of Pigati and Stern [16], yields

$$\|d\xi_\varepsilon\|_{L^2(\Omega')} + \|\varepsilon^{-1}\xi_\varepsilon\|_{L^2(\Omega')} \leq C, \quad \Omega' \Subset \Omega.$$

Passing from this bound to vanishing requires a further argument. The scalar field equation identifies the weak limit of  $\varepsilon^{-2}h_\varepsilon d\text{vol}_g$  with the energy measure. The non-negative sequence  $\varepsilon^{-1}\xi_\varepsilon^-$  is bounded in  $L^2$  and satisfies  $\varepsilon^{-1}\xi_\varepsilon^- \leq \varepsilon^{-2}h_\varepsilon$ ; any of its weak  $L^2$  limits must therefore be absolutely continuous and dominated by a measure

concentrated on a one-dimensional set. It follows that  $\varepsilon^{-1}\xi_\varepsilon \rightarrow 0$  in  $L^1_{\text{loc}}$ , after including the separate positive-part estimate. Quantitative control thus replaces exact vanishing at fixed  $\varepsilon$ , and vanishing is recovered in the singular limit.

In three dimensions, the curvature also determines a vector field and its unit direction,

$$b_\varepsilon = (*F_{\nabla_\varepsilon})^\sharp, \quad \nu_\varepsilon = \frac{b_\varepsilon}{|b_\varepsilon|} \quad \text{where } F_{\nabla_\varepsilon} \neq 0.$$

Here the Hodge star and the metric identify two-forms with vectors. The same stability calculation retains a second nonnegative term which, combined with the discrepancy derivative bound and an identity for  $b_\varepsilon$ , gives

$$\int_{\Omega' \cap \{|F_{\nabla_\varepsilon}| > 0\}} \varepsilon^2 |F_{\nabla_\varepsilon}|^2 |\nabla_{\nu_\varepsilon} \nu_\varepsilon|^2 d\text{vol}_g \leq C.$$

This controls the bending of the magnetic direction with the curvature part of the energy as weight. Further defect estimates follow by testing the discrepancy equation against a cutoff times  $h_\varepsilon$ . These deductions use the discrepancy bounds already obtained, so the analytic estimates share the same localized stability input.

To use these estimates in the limit, we use the modified Jacobian whose associated vector field  $q_\varepsilon$  is divergence-free and supported in a fixed sublevel set  $\{|u_\varepsilon|^2 \leq b_0\}$ , with  $b_0 < 1$  (see [5, III.25]). Although  $b_\varepsilon$  is already divergence-free, localization near the vortex cores is needed to compare the curvature weight with the energy density. A correction to the cutoff magnetic field preserves zero divergence. The bending and defect estimates then give the local bound

$$\limsup_{\varepsilon \rightarrow 0} \int_{B_r(x)} |\nabla_{q_\varepsilon} \nu_\varepsilon| d\text{vol}_g \leq C\sqrt{r}.$$

In particular, the contribution of this derivative becomes small on small balls, uniformly after passage to the limit.

The use of Jacobians and their relation to the limiting geometry builds on existing work. Pigati–Stern compactness includes identification of the limiting stress tensor with the tangent projection of  $V$ . Jacobians localized near vortex cores also appear in their work on the nonmagnetic Ginzburg–Landau model [17]. Related energy–Jacobian compatibility near multiplicity-one planar limits is developed by

De Philippis, Halavati, and Pigati [8]. We verify the mass and direction compatibility needed for our choice of  $q_\varepsilon$ :  $|q_\varepsilon| d\text{vol}_g \rightharpoonup 2\pi\|V\|$ , and the magnetic direction, weighted by energy, converges to the unoriented tangent of  $V$ . These are unsigned statements and are consistent with cancellation in an oriented current limit. The additional control needed for the present argument is the quantitative turning estimate and its use to exclude branching.

The final step converts this estimate into a condition on each vertex. The equation  $\text{div } q_\varepsilon = 0$  permits integration by parts with test functions depending on both position and the direction  $\nu_\varepsilon$ . The mass and direction identification passes the position derivative to  $V$ , while the turning estimate controls the term containing the direction derivative. Testing near a vertex with a radial cutoff times an arbitrary smooth odd function  $a$  of the direction gives, first letting  $\varepsilon \rightarrow 0$  and then shrinking the ball,

$$\left| \sum_i m_i a(v_i) \right| \leq C_a \sqrt{r}, \quad \text{hence} \quad \sum_i m_i a(v_i) = 0.$$

The radial cutoff contributes a fixed endpoint term on each arm, whereas the test function's derivative in the direction variable stays bounded as  $r \downarrow 0$ . Unlike stationarity, which tests only linear functions of the direction, this identity holds for every odd  $a$ . It therefore implies

$$\sum_i m_i \delta_{v_i} = \sum_i m_i \delta_{-v_i}.$$

Each arm pairs with an opposite arm of equal total multiplicity. Uniqueness for the geodesic equation joins these pairs into full smooth geodesics, proving the stated regularity of the limit.

*Remark 1.1.* No better regularity than the one predicted by Theorem A can hold for limits of stable critical points, as shown by the following example. Consider the Riemannian manifold  $(\mathbf{T}^3, g)$  with the metric  $g = (1 + \sin^2 y + V(z))dx + (1 + \sin^2 x + V(z))dy + dz$ ,

$$V(z) = \begin{cases} e^{-1/z^2} \sin^2(1/z), & z \neq 0, \\ 0, & z = 0. \end{cases}$$

consider the disjoint curves  $\gamma_{1,j}(s) = (s, 0, a_j)$  and  $\gamma_{2,j}(s) = (0, s, -a_j)$ , where  $a_j = (\pi j)^{-1}$ . Each pair is a strict local minimum of length, so the abelian Higgs  $\Gamma$ -limit [13] provides stable local minimizers concentrating on it as  $\varepsilon \rightarrow 0$ . A diagonal choice  $\varepsilon_j \ll a_j$  yields a limiting pair of closed geodesics crossing transversely at the origin.

## 2. PRELIMINARIES AND THE STABILITY CALCULATION

We denote a typical section–real one-form pair by  $\Phi = (\phi, \Phi)$ , and reserve  $\mathbf{U} = (u, \nabla)$  for the background solution. Throughout the remainder of the paper,  $\dim M = 3$  and  $|u| \leq 1$ . Angled brackets denote the real fibre inner product, given in a unitary trivialisation by  $\langle a, b \rangle = \operatorname{Re}(a\bar{b})$ . Locally we write  $\nabla^A = d - iA$ , where  $A$  is a real one-form; then  $F_\nabla = dA$ . We also use  $\nabla$  for the induced connections on tensor-valued sections and real tensors. We use the convention that the Hodge Laplacian has nonpositive spectrum  $\Delta := -(dd^* + d^*d)$ . If  $F$  is the skew-symmetric matrix representing  $F_\nabla$  in an orthonormal frame, then

$$|F_\nabla|^2 = \frac{1}{2} \sum_{i,j} F_{ij}^2, \quad \|F\|^2 = 2|F_\nabla|^2,$$

where  $\|\cdot\|$  denotes the Hilbert–Schmidt norm.

Set

$$h(u) := \frac{1}{2}(1 - |u|^2), \quad j_\nabla u := \langle \nabla u, iu \rangle.$$

When no confusion is possible, we write  $h = h(u)$ . The Euler–Lagrange equations of (1.1) are

$$\nabla^* \nabla u = \varepsilon^{-2} h(u) u, \quad \varepsilon^2 d^* F_\nabla = j_\nabla u.$$

By the assumption  $|u| \leq 1$ , we have  $0 \leq h \leq \frac{1}{2}$ . Define the real two-form

$$K(X, Y) := -2 \langle \nabla_X u, i \nabla_Y u \rangle.$$

Direct differentiation and [16, (2.7)] give

$$d(j_\nabla u) = K - |u|^2 F_\nabla, \quad d|u|^2 \wedge j_\nabla u = |u|^2 K, \quad |K| \leq |\nabla u|^2. \quad (2.1)$$

The scalar and curvature equations take the form [16, (3.4), (2.6)]

$$(-\varepsilon^2 \Delta + |u|^2) h = \varepsilon^2 |\nabla u|^2, \quad (-\varepsilon^2 \Delta + |u|^2) F_\nabla = K. \quad (2.2)$$

**2.1. Discrepancy, interior estimates and compactness.** Here we collect the interior estimates and compactness results needed below, mainly from [16]. Fix  $\Omega \subset M^3$  and  $\Omega' \Subset \Omega$ , and consider smooth critical points with  $|u| \leq 1$  and  $E_\varepsilon(u, \nabla; \Omega) \leq \Lambda$ .

The key to the  $(n - 2)$ -monotonicity of the energy, and consequent compactness, is the upper bound on the discrepancy

$$\xi_\varepsilon := \varepsilon |F_\nabla| - \varepsilon^{-1} h(u). \quad (2.3)$$

We write  $\xi_\varepsilon^\pm = \max\{\pm \xi_\varepsilon, 0\}$  for its positive and negative parts. The required upper bound was proved in [16, Proposition 4.2]. We record the following interior versions of [16, Proposition 4.2, Theorem 4.3 and Proposition 5.1], together with the curvature derivative estimate. For sufficiently small  $\varepsilon$ , there are constants  $C, r_0 > 0$ , depending only on the domains, their geometry and  $\Lambda$ , such that on  $\Omega'$

$$\xi_\varepsilon \leq C, \quad (2.4)$$

$$|\nabla u|^2 \leq C\varepsilon^{-2}h^2 + C\varepsilon^2, \quad \varepsilon^2|F_\nabla|^2 \leq C\varepsilon^{-2}h^2 + C\varepsilon, \quad (2.5)$$

$$|\nabla F_\nabla| \leq C\varepsilon^{-3}, \quad (2.6)$$

and

$$E_\varepsilon(u, \nabla; B_r(x)) \leq Cr \quad (x \in \Omega', \ 0 < r < r_0). \quad (2.7)$$

*Justification of the interior estimates.* All estimates are obtained on successive fixed domains compactly contained in  $\Omega$ . The curvature equation and the two-form Weitzenböck identity give

$$-\Delta F_\nabla = \nabla^* \nabla F_\nabla + \mathcal{R}_2 F_\nabla,$$

where  $\mathcal{R}_2$  is the curvature endomorphism of two-forms. The discrepancy inequality [16, (3.7)] and the positive-part chain rule imply

$$\Delta \xi_\varepsilon^+ \geq -C\varepsilon |F_\nabla|.$$

The energy bounds both  $\xi_\varepsilon^+$  and  $\varepsilon |F_\nabla|$  in  $L^2$ . Since  $2 > 3/2$ , the interior elliptic estimate for subsolutions gives (2.4). Together with  $h \leq 1/2$ , this yields  $|F_\nabla| \leq C\varepsilon^{-2}$  on smaller domains.

The inner-variation calculation in [16, Section 4] is local, and its discrepancy term satisfies

$$\varepsilon^2 |F_\nabla|^2 - \varepsilon^{-2} h^2 = \xi_\varepsilon(\varepsilon |F_\nabla| + \varepsilon^{-1} h) \leq C e_\varepsilon^{1/2}.$$

The monotonicity argument of [16, Theorem 4.3], applied inside a fixed larger domain, therefore gives (2.7). The Bochner inequality [16, (5.4)] implies  $\Delta |\nabla u|^2 \geq -C\varepsilon^{-2} |\nabla u|^2$ . The interior mean-value estimate on  $B_\varepsilon(x)$  and (2.7) yield

$$|\nabla u|^2(x) \leq C\varepsilon^{-3} E_\varepsilon(u, \nabla; B_\varepsilon(x)) \leq C\varepsilon^{-2}.$$

On rescaled  $\varepsilon$ -balls,  $\varepsilon^2 F_\nabla$  and the source  $\varepsilon^2 K$  are uniformly bounded. Interior  $W^{2,q}$  estimates for the curvature equation, with  $q > 3$ , then give (2.6).

To localise the sharper bounds in (2.5), set  $v = |\nabla u| - 2\varepsilon^{-1} h$ . The calculation in [16, (5.5)–(5.6)] gives, on  $\{v > 0\}$  and for small  $\varepsilon$ ,

$$\varepsilon^2 \Delta v \geq (1 + 3h - C\varepsilon)v + 2\varepsilon v^2 + 2\varepsilon^{-1} h^2 - 2Ch \geq \frac{1}{2}v - C\varepsilon.$$

Thus  $(v - C\varepsilon)_+$  satisfies a screened subsolution inequality. Its coarse bound  $C\varepsilon^{-1}$  and comparison with exponential barriers on fixed normal balls give  $v \leq C\varepsilon$  on smaller domains, proving the gradient bound in (2.5). Similarly, the calculation in [16, proof of Proposition 5.1] shows that the positive part of

$$\varepsilon |F_\nabla| - (1 + C\varepsilon)\varepsilon^{-1} h - C\sqrt{\varepsilon}$$

satisfies a screened subsolution inequality at scale  $\varepsilon^{3/4}$ . The same interior comparison, as in [16, proof of Proposition 5.3], gives  $\varepsilon |F_\nabla| \leq (1 + C\varepsilon)\varepsilon^{-1} h + C\sqrt{\varepsilon}$  on smaller domains. Squaring proves the curvature bound in (2.5).  $\square$

We also record the localised version of the compactness statement in dimension three.

**Theorem 2.1** (Pigati–Stern). *Let  $\varepsilon_j \rightarrow 0$  and let  $(u_j, \nabla_j)$  be smooth critical points in  $\Omega$  with  $|u_j| \leq 1$  and  $E_{\varepsilon_j}(u_j, \nabla_j; \Omega) \leq \Lambda$ . After passing to a subsequence, there is a stationary integral one-varifold  $V$  in  $\Omega$  such that*

$$e_{\varepsilon_j}(u_j, \nabla_j) d\text{vol}_g \rightharpoonup \mu := 2\pi \|V\|$$



as locally finite measures. In particular,  $\mu$  is singular with respect to volume. If  $\tau$  is either unit orientation of its tangent line, defined  $\mu$ -almost everywhere, then

$$T_{\varepsilon_j} d\text{vol}_g \rightharpoonup \tau^\flat \otimes \tau^\flat \mu, \quad T_\varepsilon := e_\varepsilon g - 2S - 2\varepsilon^2 F^\top F, \quad (2.8)$$

where  $S(X, Y) := \langle \nabla_X u, \nabla_Y u \rangle$ .

*Proof.* This is the interior form of [16, Theorem 1.1 and Proposition 6.4], with the normalisation (1.1). Once the preceding interior estimates are available, the clearing-out, decay, compactness and integrality arguments in Sections 4–6 of that paper apply on successively smaller domains. In the decay argument, the distance to the vortex region is truncated by the distance to the boundary of a fixed larger interior domain. Exhaustion and a diagonal subsequence give the stated local convergence.  $\square$

**2.2. Scalar coercivity.** The following coercivity estimate follows from the local energy bound, which ensures that  $|u|^2 \geq 1/2$  on a fixed positive proportion of every ball of radius  $R\varepsilon$ , for a sufficiently large fixed  $R$ .

**Lemma 2.2.** *For every  $\Omega' \Subset \Omega$ , all sufficiently small  $\varepsilon$  and every real  $z \in H_0^1(\Omega')$ , extended by zero to  $\Omega$ ,*

$$\int_{\Omega} z^2 d\text{vol}_g \leq C \int_{\Omega} (\varepsilon^2 |dz|^2 + |u|^2 z^2) d\text{vol}_g.$$

*Proof.* On  $\{|u|^2 < \frac{1}{2}\}$  the potential energy density is bounded below by  $\frac{1}{16}\varepsilon^{-2}$ . For a fixed  $R > 1$ , (2.7) therefore gives

$$\text{vol}_g(B_{R\varepsilon}(x) \cap \{|u|^2 < 1/2\}) \leq C\varepsilon^2 E_\varepsilon(u, \nabla; B_{R\varepsilon}(x)) \leq CR\varepsilon^3.$$

On the other hand we have the volume lower bound  $\text{vol}_g(B_{R\varepsilon}(x)) \geq cR^3\varepsilon^3$ , for some  $c > 0$ . Thus, we can pick  $R$  such that, denoting  $G := B_{R\varepsilon}(x) \cap \{|u|^2 \geq \frac{1}{2}\}$  and  $B := B_{R\varepsilon}(x)$ , we have

$$\text{vol}_g(G) \geq \frac{1}{2} \text{vol}_g(B) \quad (2.9)$$

uniformly in  $\varepsilon$  and  $x \in \Omega'$ .

Let  $z_B$  be the average of  $z$  on  $B$ . The lower bound (2.9) implies that

$$\text{vol}_g(B)|z_B|^2 \leq 2\text{vol}_g(G)|z_B|^2 \leq 4 \int_G z^2 + 4 \int_G |z - z_B|^2$$

Thus, by Poincaré's inequality and the fact that  $1 \leq 2|u|^2$  on  $G$ ,

$$\begin{aligned} \int_B z^2 d\text{vol}_g &= \int_B |z - z_B|^2 + \text{vol}_g(B) |z_B|^2 \\ &\leq 5 \int_B |z - z_B|^2 + 4 \int_G |z|^2 \\ &\leq CR^2 \varepsilon^2 \int_B |dz|^2 + C \int_B |u|^2 |z|^2 \end{aligned}$$

Since  $R$  is fixed and uniform in  $\varepsilon$ , it can be absorbed in the constant. A covering argument concludes the proof.  $\square$

**2.3. Stability and covariant test pairs.** Here we adapt the stability tests used in [2] to the manifold setting, which leaves a geometrically controlled error.

Recall that the second variation of (1.1) is

$$\begin{aligned} Q_{\varepsilon, \mathbf{U}}[\Phi] &= \int_{\Omega} (|\nabla \phi - i\Phi u|^2 - 2\langle \nabla u, i\Phi \phi \rangle + \varepsilon^2 |d\Phi|^2 \\ &\quad + \varepsilon^{-2} \langle u, \phi \rangle^2 - \varepsilon^{-2} h(u) |\phi|^2) d\text{vol}_g. \end{aligned}$$

Stability of  $\mathbf{U}$  means  $Q_{\varepsilon, \mathbf{U}}[\Phi] \geq 0$  for every smooth compactly supported pair  $\Phi$ . It is natural to consider the gauge-corrected form

$$\begin{aligned} \mathbf{Q}_{\varepsilon, \mathbf{U}}[\Phi] &:= \int_{\Omega} (|\nabla \phi|^2 + \varepsilon^2 |d\Phi|^2 + \varepsilon^2 |d^* \Phi|^2 + 4\langle i\Phi \cdot \nabla u, \phi \rangle \\ &\quad + \varepsilon^{-2} (|u|^2 - h(u)) |\phi|^2 + |u|^2 |\Phi|^2) d\text{vol}_g, \end{aligned}$$

which satisfies the exact identity

$$\begin{aligned} \mathbf{Q}_{\varepsilon, \mathbf{U}}[\Phi] &= Q_{\varepsilon, \mathbf{U}}[\Phi] + \varepsilon^2 \int_{\Omega} G_{\varepsilon, \mathbf{U}}[\Phi]^2 d\text{vol}_g, \\ G_{\varepsilon, \mathbf{U}}[\Phi] &:= d^* \Phi + \varepsilon^{-2} \langle iu, \phi \rangle. \end{aligned} \tag{2.10}$$

We denote the bilinear form associated to  $Q_{\varepsilon, \mathbf{U}}$  by  $\mathcal{B}_{\varepsilon, \mathbf{U}}$  and the gauge-corrected linearised operator by  $L_{\varepsilon, \mathbf{U}}$ , as in [2]. They are related by

$$\int_{\Omega} L_{\varepsilon, \mathbf{U}}[\Phi] \cdot \Psi d\text{vol}_g = \mathcal{B}_{\varepsilon, \mathbf{U}}[\Phi, \Psi] + \varepsilon^2 \int_{\Omega} G_{\varepsilon, \mathbf{U}}[\Phi] G_{\varepsilon, \mathbf{U}}[\Psi] d\text{vol}_g.$$

All the formulas above extend to compactly supported  $H^1$  pairs by density. For the rest of the proof we omit  $\varepsilon, \mathbf{U}$  from  $Q, \mathbf{Q}, \mathcal{B}, G$  and  $L$ .

On a neighbourhood of the compact supports under consideration, choose a smooth isometric embedding  $\iota$  into  $\mathbf{R}^N$  and set  $X_a = \nabla \iota^a$ . The projected coordinate fields satisfy

$$\sum_a X_a \otimes X_a = g^{-1}, \quad \sum_a X_a \otimes \nabla_Y X_a = 0 \quad (2.11)$$

for every vector field  $Y$ . The first identity expresses isometry. For the second, let  $E_a$  be the standard basis of  $\mathbf{R}^N$  and  $\Pi$  the second fundamental form of  $\iota$ . For vector fields  $Z, W$ ,

$$\langle X_a, Z \rangle = \langle d\iota(Z), E_a \rangle, \quad \langle \nabla_Y X_a, W \rangle = \nabla^2 \iota^a(Y, W) = \langle \Pi(Y, W), E_a \rangle,$$

and hence

$$\sum_a \langle X_a, Z \rangle \langle \nabla_Y X_a, W \rangle = \langle d\iota(Z), \Pi(Y, W) \rangle = 0.$$

Define the fixed symmetric tensor  $H \in \Gamma(\text{Sym}^2 T^*M)$  by

$$H_{ij} := \sum_{a,k} \langle \nabla_{e_k} X_a, e_i \rangle \langle \nabla_{e_k} X_a, e_j \rangle.$$

**Lemma 2.3.** *For every compactly supported  $T \in H^1(\Omega; T^*M \otimes T^*M)$ ,*

$$\begin{aligned} \sum_a \int_{\Omega} (|d(T(X_a, \cdot))|^2 + |d^*(T(X_a, \cdot))|^2) d\text{vol}_g &= \int_{\Omega} |\nabla T|^2 d\text{vol}_g \\ &+ \int_{\Omega} \sum_i H(T(\cdot, e_i)^\sharp, T(\cdot, e_i)^\sharp) d\text{vol}_g \\ &+ \int_{\Omega} \sum_i \text{Ric}(T(e_i, \cdot)^\sharp, T(e_i, \cdot)^\sharp) d\text{vol}_g, \end{aligned} \quad (2.12)$$

where  $\{e_i\}$  is any local orthonormal frame.

*Proof.* For smooth compactly supported  $T$ , set  $t_a = T(X_a, \cdot)$ . The one-form Weitzenböck identity [15, Theorem 9.4.1], integrated over  $\Omega$ , gives

$$\int_{\Omega} (|dt_a|^2 + |d^*t_a|^2) d\text{vol}_g = \int_{\Omega} (|\nabla t_a|^2 + \text{Ric}(t_a^\sharp, t_a^\sharp)) d\text{vol}_g.$$

Expand  $\nabla t_a = (\nabla T)(X_a, \cdot) + T(\nabla X_a, \cdot)$  and sum in  $a$ . The mixed term vanishes by (2.11), so

$$\begin{aligned} \sum_a |\nabla t_a|^2 &= |\nabla T|^2 + \sum_i H(T(\cdot, e_i)^\sharp, T(\cdot, e_i)^\sharp), \\ \sum_a \text{Ric}(t_a^\sharp, t_a^\sharp) &= \sum_i \text{Ric}(T(e_i, \cdot)^\sharp, T(e_i, \cdot)^\sharp). \end{aligned}$$

Approximation gives the identity for  $H^1$  tensors.  $\square$

Now, set

$$b := (*F_\nabla)^\sharp, \quad \nu := \frac{b}{|b|} \quad \text{on } \{|F_\nabla| > 0\}, \quad B := \sqrt{-F^2}.$$

In dimension three,

$$B = |F_\nabla|(g - \nu^\flat \otimes \nu^\flat), \quad FF^\top = F^\top F = B^2, \quad \|\nabla F\|^2 = \|\nabla B\|^2. \quad (2.13)$$

We identify  $B$  with the symmetric bilinear form obtained by lowering an index with  $g$ , and write  $B^2(X, Y) = \langle B(X, \cdot), B(Y, \cdot) \rangle$ . We extend  $B$  by zero on  $\{|F_\nabla| = 0\}$ . The map  $b \mapsto |b|I - b \otimes \frac{b}{|b|}$ , with this extension, is Lipschitz, so  $B \in H_{\text{loc}}^1(\Omega; \text{Sym}^2 T^*M)$ .

We write  $\dagger$  for the conjugate transpose in an orthonormal frame. As in [2], consider the pairs

$$\mathbf{V}_a := (\nabla_{X_a} u, \iota_{X_a} F_\nabla), \quad \mathbf{W}_a := (i\nabla_{X_a} u, -B(X_a, \cdot)).$$

**Proposition 2.4.** *For every real  $\zeta \in C_c^\infty(\Omega)$ ,*

$$\sum_a \mathbf{Q}[\zeta \mathbf{W}_a] + 4 \int_\Omega \zeta^2 (\nabla u)^\dagger (B + iF) (\nabla u) d\text{vol}_g = \sum_a \mathbf{Q}[\zeta \mathbf{V}_a]. \quad (2.14)$$

Moreover, the Hermitian endomorphism  $B + iF$  is positive semidefinite, thus if the critical point is stable in  $\Omega$ , then

$$0 \leq \sum_a \mathbf{Q}[\zeta \mathbf{W}_a] \leq C \int_\Omega (\zeta^2 + |d\zeta|^2) e_\varepsilon(u, \nabla) d\text{vol}_g. \quad (2.15)$$

*Proof.* Apply the calculations of [2, Section 3.1, equations (3.5)–(3.8)] and Lemma 2.3 to the compactly supported covariant tensors  $\zeta F_\nabla$  and  $\zeta B$ . By (2.13), they have the same norms, derivative norms and geometric contributions. The difference of the mixed terms is  $-4\zeta^2 (\nabla u)^\dagger (B + iF) (\nabla u)$ , proving (2.14).

To estimate the right-hand side, use the Bochner calculations for  $\nabla u$  and  $F_\nabla$  in [16, Section 3, equation (3.1) and the calculation following (3.4)]. The terms involving  $\langle F_\nabla, K \rangle$  cancel. The scalar Laplacian term is

$$\frac{1}{2} \int_{\Omega} \zeta^2 \Delta (|\nabla u|^2 + 2\varepsilon^2 |F_\nabla|^2) d\text{vol}_g,$$

which cancels the mixed cutoff term after integration by parts. Consequently,

$$\begin{aligned} \sum_a \mathbf{Q}[\zeta \mathbf{V}_a] &= \int_{\Omega} \zeta^2 (\langle H - \text{Ric}, S \rangle + \varepsilon^2 \langle H + \text{Ric}, B^2 \rangle \\ &\quad - 2\varepsilon^2 \langle \mathcal{R}_2 F_\nabla, F_\nabla \rangle) d\text{vol}_g \\ &\quad + \int_{\Omega} |d\zeta|^2 (|\nabla u|^2 + 2\varepsilon^2 |F_\nabla|^2) d\text{vol}_g. \end{aligned}$$

Finally,  $B + iF = |iF| + iF$  is positive semidefinite. Stability for the compactly supported pairs  $\zeta \mathbf{W}_a$  and boundedness of the fixed geometric tensors give (2.15).  $\square$

### 3. THE DISCREPANCY BOUND

Recall Pigati–Stern’s discrepancy  $\xi_\varepsilon$  from (2.3). In the notation of [2, Section 3.1],

$$-\varepsilon \xi_\varepsilon = h(u) - \frac{\varepsilon^2}{2} \text{tr } B = h(u) - \varepsilon^2 |F_\nabla|.$$

The pointwise estimate (2.4) from [16] controls the positive part  $\xi_\varepsilon^+$ . Here we show that in dimension three *stability* controls the negative part  $\xi_\varepsilon^-$ , including its derivative, in an integral way.

**Theorem 3.1** (Interior discrepancy estimate). *For every smooth critical point in  $\Omega$  with  $|u| \leq 1$  and  $E_\varepsilon(u, \nabla; \Omega) \leq \Lambda$ , every  $\Omega' \Subset \Omega$ , and all sufficiently small  $\varepsilon$ ,*

$$\|\varepsilon^{-1} \xi_\varepsilon^+\|_{L^2(\Omega')} + \|d\xi_\varepsilon^+\|_{L^2(\Omega')} \leq C\varepsilon. \quad (3.1)$$

*If the critical point is stable with respect to compactly supported variations in  $\Omega$ , then*

$$\|\varepsilon^{-1} \xi_\varepsilon\|_{L^2(\Omega')} + \|d\xi_\varepsilon\|_{L^2(\Omega')} \leq C. \quad (3.2)$$

*Moreover, for a sequence of stable critical points whose energy measures converge locally to  $\mu$  as in Theorem 2.1,*

$$\|\varepsilon^{-1} \xi_\varepsilon\|_{L^1(\Omega')} \longrightarrow 0, \quad (3.3)$$

and, as locally finite measures in  $\Omega$ ,

$$\varepsilon^{-2} h \, d\text{vol}_g \rightharpoonup \mu, \quad |F_\nabla| \, d\text{vol}_g \rightharpoonup \mu. \quad (3.4)$$

*Proof.* First, we show that every critical point satisfies (3.1), which is just a consequence of the discrepancy bound by [16]. Indeed, by [16, Equation (3.7)], we have

$$(-\varepsilon^2 \Delta + |u|^2) \xi_\varepsilon \leq C \varepsilon^3 |F_\nabla|.$$

For  $\eta \in C_c^\infty(\Omega)$ , testing against  $\eta^2 \xi_\varepsilon^+$  gives

$$\begin{aligned} \int_{\Omega} (\varepsilon^2 |d(\eta \xi_\varepsilon^+)|^2 + |u|^2 \eta^2 (\xi_\varepsilon^+)^2) \, d\text{vol}_g &\leq C \varepsilon^3 \int_{\Omega} \eta^2 |F_\nabla| \xi_\varepsilon^+ \, d\text{vol}_g \\ &\quad + \varepsilon^2 \int_{\Omega} |d\eta|^2 (\xi_\varepsilon^+)^2 \, d\text{vol}_g. \end{aligned}$$

Lemma 2.2, the energy bound and Young's inequality yield

$$\|\eta \xi_\varepsilon^+\|_{L^2}^2 + \varepsilon^2 \|d(\eta \xi_\varepsilon^+)\|_{L^2}^2 \leq C \varepsilon^4 + C \varepsilon^2 \int_{\Omega} |d\eta|^2 (\xi_\varepsilon^+)^2 \, d\text{vol}_g.$$

Initially, the energy bound gives  $\|\xi_\varepsilon^+\|_{L^2(\Omega)}^2 \leq C$ . Apply the preceding inequality with two nested cutoffs, the inner one equal to one on  $\Omega'$  and supported where the outer one is one. The first application gives an interior squared  $L^2$  bound  $C \varepsilon^2$  for  $\xi_\varepsilon^+$ , and the second gives  $C \varepsilon^4$ , together with  $\|d\xi_\varepsilon^+\|_{L^2(\Omega')}^2 \leq C \varepsilon^2$ . This proves (3.1).

Next, we assume that the critical points  $(u, \nabla)$  are stable. The proof of (3.2) can be understood as a quantitative version of the estimates in [2].

For every real  $\psi \in H^1(\Omega)$  with compact support, set  $\Psi_a = (0, \psi X_a^b)$ . Then, using the polarisation of (2.12), we find

$$\begin{aligned} \sum_a \int_{\Omega} L[\mathbf{W}_a] \cdot \Psi_a \, d\text{vol}_g &= -\varepsilon^2 \int_{\Omega} \langle d(\text{tr } B), d\psi \rangle \, d\text{vol}_g \\ &\quad + \int_{\Omega} \psi (2|\nabla u|^2 - |u|^2 \text{tr } B) \, d\text{vol}_g \\ &\quad - \varepsilon^2 \int_{\Omega} \psi \langle H + \text{Ric}, B \rangle \, d\text{vol}_g. \end{aligned}$$

Testing the scalar equation in (2.2) against  $2\psi$  and using that  $2h - \varepsilon^2 \operatorname{tr} B = -2\varepsilon\xi_\varepsilon$ , we get

$$\begin{aligned} \sum_a \int_\Omega L[\mathbf{W}_a] \cdot \Psi_a \, d\operatorname{vol}_g &= -2\varepsilon \int_\Omega (\langle d\xi_\varepsilon, d\psi \rangle + \varepsilon^{-2}|u|^2\xi_\varepsilon\psi) \, d\operatorname{vol}_g \\ &\quad - \varepsilon^2 \int_\Omega \psi \langle H + \operatorname{Ric}, B \rangle \, d\operatorname{vol}_g. \end{aligned} \quad (3.5)$$

Moreover, by applying Lemma 2.3 to  $T = \psi g$ , we get

$$\sum_a \mathbf{Q}[\Psi_a] = \int_\Omega (3\varepsilon^2|d\psi|^2 + 3|u|^2\psi^2 + \varepsilon^2(\operatorname{tr} H + \operatorname{Scal})\psi^2) \, d\operatorname{vol}_g. \quad (3.6)$$

Choose  $0 \leq \eta, \zeta \leq 1$  in  $C_c^\infty(\Omega)$ , with  $\eta = 1$  on  $\Omega'$  and  $\zeta = 1$  on a neighbourhood of  $\operatorname{supp} \eta$ . Set  $\psi = -\varepsilon^{-1}\eta^2\xi_\varepsilon$ . The pairing of  $\zeta\mathbf{W}_a$  with  $\Psi_a$  equals that of  $\mathbf{W}_a$  with  $\Psi_a$ , since  $d\zeta = 0$  near its support. By (2.15), Cauchy–Schwarz for  $\mathbf{Q}$ , (3.6) and Lemma 2.2,

$$\begin{aligned} \left| \sum_a \int_\Omega L[\mathbf{W}_a] \cdot \Psi_a \, d\operatorname{vol}_g \right| &\leq \left( \sum_a \mathbf{Q}[\zeta\mathbf{W}_a] \right)^{1/2} \left( \sum_a \mathbf{Q}[\Psi_a] \right)^{1/2} \\ &\leq C(\|d(\eta\xi_\varepsilon)\|_{L^2} + \|\varepsilon^{-1}|u|\eta\xi_\varepsilon\|_{L^2}). \end{aligned}$$

The geometric term in (3.5) is bounded by

$$\begin{aligned} C\varepsilon \int_\Omega \eta^2 |F_\nabla| |\xi_\varepsilon| \, d\operatorname{vol}_g &\leq C\varepsilon \|\varepsilon^{-1}\eta\xi_\varepsilon\|_{L^2} \\ &\leq C\varepsilon (\|d(\eta\xi_\varepsilon)\|_{L^2} + \|\varepsilon^{-1}|u|\eta\xi_\varepsilon\|_{L^2}). \end{aligned}$$

The energy bound gives  $\int_\Omega |d\eta|^2 |\xi_\varepsilon|^2 \, d\operatorname{vol}_g \leq C$ . Using

$$\langle d\xi_\varepsilon, d(\eta^2\xi_\varepsilon) \rangle = |d(\eta\xi_\varepsilon)|^2 - |d\eta|^2 (\xi_\varepsilon)^2,$$

equation (3.5) therefore yields

$$\|d(\eta\xi_\varepsilon)\|_{L^2}^2 + \|\varepsilon^{-1}|u|\eta\xi_\varepsilon\|_{L^2}^2 \leq C(\|d(\eta\xi_\varepsilon)\|_{L^2} + \|\varepsilon^{-1}|u|\eta\xi_\varepsilon\|_{L^2}) + C.$$

Both norms are bounded. Lemma 2.2 supplies the unweighted  $L^2$  bound for  $\varepsilon^{-1}\eta\xi_\varepsilon$ , and  $\eta = 1$  on  $\Omega'$  proves (3.2).

Lastly, we prove (3.3) and (3.4). Equation (2.2) gives, for every  $0 \leq \chi \in C_c^\infty(\Omega)$ ,

$$\int_{\Omega} \chi \varepsilon^{-2} h \, d\text{vol}_g = \int_{\Omega} \chi (|\nabla u|^2 + 2\varepsilon^{-2} h^2) \, d\text{vol}_g + \int_{\Omega} h \Delta \chi \, d\text{vol}_g \leq C_\chi.$$

Since  $h \geq 0$ , the measures  $\varepsilon^{-2} h \, d\text{vol}_g$  have locally bounded mass. In particular,  $h \rightarrow 0$  in  $L_{\text{loc}}^1$ , so  $\Delta h \rightarrow 0$  distributionally. The same equation also gives

$$e_\varepsilon - \varepsilon^{-2} h = -\Delta h + \varepsilon^{-1} \xi_\varepsilon (h + \varepsilon^2 |F_\nabla|).$$

By (3.2),  $\varepsilon^{-1} \xi_\varepsilon$  is bounded in  $L_{\text{loc}}^2$ , while  $\|h + \varepsilon^2 |F_\nabla|\|_{L^2(\Omega)} \leq C\varepsilon$  by the energy bound, hence the last term tends to zero in  $L_{\text{loc}}^1$ . Hence  $\varepsilon^{-2} h \, d\text{vol}_g \rightarrow \mu$  locally.

The negative part  $\varepsilon^{-1} \xi_\varepsilon^-$  is locally bounded in  $L^2$  and satisfies  $0 \leq \varepsilon^{-1} \xi_\varepsilon^- \leq \varepsilon^{-2} h$ . Every weak  $L^2$  limit on an interior open set therefore defines an absolutely continuous nonnegative measure dominated by  $\mu$ . Since  $\mu$  is singular with respect to volume, the  $L^2$  limit is zero. Testing weak  $L^2$  convergence against the indicator of  $\Omega'$  gives  $\|\varepsilon^{-1} \xi_\varepsilon^-\|_{L^1(\Omega')} \rightarrow 0$ . The rescaled positive part  $\varepsilon^{-1} \xi_\varepsilon^+$  tends to zero in  $L^2(\Omega')$  by (3.1), proving (3.3). Finally,  $|F_\nabla| = \varepsilon^{-2} h + \varepsilon^{-1} \xi_\varepsilon$  gives (3.4).  $\square$

#### 4. GEOMETRIC CONSEQUENCES

Fix a sequence of smooth critical points with  $\varepsilon \rightarrow 0$ ,  $|u| \leq 1$  and uniformly bounded energy, stable under compactly supported variations in  $\Omega$ . Pass to a subsequence defining  $\mu = 2\pi\|V\|$  as in Theorem 2.1. For the estimates below, we introduce a modified Jacobian, as in [6, (III.23)] and [17, Section 2].

Fix  $b_0 \in (0, 1)$  and a smooth nondecreasing function  $t: [0, \infty) \rightarrow [0, 1]$  which is zero near zero and equals one on  $[b_0, \infty)$ . Define  $\chi(a) = t(a)/a$ , extended smoothly by zero near  $a = 0$ , and set

$$J_\varepsilon := F_\nabla + d(\chi(|u|^2) j_\nabla u), \quad q_\varepsilon := (*J_\varepsilon)^\sharp.$$

By (2.1),

$$J_\varepsilon = (1 - t(|u|^2)) F_\nabla + t'(|u|^2) K. \tag{4.1}$$

Consequently,

$$dJ_\varepsilon = 0, \quad \text{div } q_\varepsilon = 0, \quad \text{spt } q_\varepsilon \subset \{|u|^2 \leq b_0\}.$$



On this core,  $h \geq c_0 := (1 - b_0)/2 > 0$ . We first state the integral curvature estimate for  $q_\varepsilon$ .

**Proposition 4.1** (Integral curvature estimates). *Fix  $\Omega' \Subset \Omega$ . For all sufficiently small  $\varepsilon$ ,  $\nu$  is smooth near  $\text{spt } q_\varepsilon \cap \overline{\Omega'}$ . For every normal ball  $B_{2r}(x) \subset \Omega'$  with  $0 < 2r < r_0$ ,*

$$\int_{B_r(x)} |\nabla_{q_\varepsilon} \nu| d\text{vol}_g \leq CE_\varepsilon(u, \nabla; B_{2r}(x))^{1/2} (1 + \varepsilon/r) + C\varepsilon E_\varepsilon(u, \nabla; B_{2r}(x)). \quad (4.2)$$

The integrand is extended by zero where  $q_\varepsilon = 0$ . In particular,

$$\limsup_{\varepsilon \rightarrow 0} \int_{B_r(x)} |\nabla_{q_\varepsilon} \nu| d\text{vol}_g \leq C\sqrt{r}, \quad (4.3)$$

with constant depending on  $\Omega', \Lambda$ .

The preceding proposition controls the turning of  $\nu = (*F_\nabla)^\sharp/|F_\nabla|$  along  $q_\varepsilon$ . To relate this estimate to the limit varifold, we show that  $\nu$  recovers its tangent directions and that  $q_\varepsilon$  carries its full unsigned mass. This follows from Theorem 3.1 and the first-order identities used in [2].

**Proposition 4.2.** *For any measurable unit extension of  $\nu$  to  $\{F_\nabla = 0\}$  and every  $\Omega' \Subset \Omega$ ,*

$$\|T_\varepsilon - e_\varepsilon \nu^\flat \otimes \nu^\flat\|_{L^1(\Omega')} \rightarrow 0,$$

where  $T_\varepsilon$  is the stress-energy tensor. Decompose

$$q_\varepsilon = \alpha_\varepsilon \nu + R_\varepsilon, \quad R_\varepsilon \perp \nu. \quad (4.4)$$

Then  $|q_\varepsilon| \leq Ce_\varepsilon$ , and, locally as Radon measures in  $\Omega$ ,

$$(\alpha_\varepsilon)_+ d\text{vol}_g \rightharpoonup \mu, \quad |q_\varepsilon| d\text{vol}_g \rightharpoonup \mu.$$

Moreover, for every  $\Omega' \Subset \Omega$ ,

$$\int_{\Omega'} ((\alpha_\varepsilon)_- + |R_\varepsilon|) d\text{vol}_g \rightarrow 0.$$

The preceding two propositions imply the main result.

**Theorem 4.3.** *For the sequence and hypotheses fixed at the start of this section, the limit varifold  $V$  is locally a finite union of smooth geodesic arcs without interior*

endpoints, counted with positive integer multiplicities and possibly intersecting. If  $\Omega = M$  is closed,  $V$  is a finite union of closed immersed geodesics, counted with positive integer multiplicities.

*Proof.* Pigati–Stern compactness and [1, Section 3] give a locally finite geodesic network in  $\Omega$ . We show that its outgoing directions pair antipodally at every interior vertex. Consider  $\tau$  from (2.8). Now let  $S\Omega$  be the unit tangent bundle over  $\Omega$ , and let  $\Xi(x, v) \in T_x^*M$  be continuous on  $S\Omega$ , compactly supported in the base variable, and odd in  $v$ . Then we claim:

$$\lim_{\varepsilon \rightarrow 0} \int_{\Omega} \Xi(x, \nu)(q_{\varepsilon}) d\text{vol}_g = \int_{\Omega} \Xi(x, \tau)(\tau) d\mu.$$

On the bundle  $\mathbb{P}(T\Omega)$  of unoriented unit directions, consider

$$e_{\varepsilon}(x) d\text{vol}_g(x) \delta_{[\nu(x)]}.$$

These measures have uniformly bounded mass over compact subsets of the base. A subsequential local weak limit has base measure  $\mu$ . Disintegrate it as  $\mu(dx)\sigma_x(d[v])$ , where  $\sigma_x$  is a probability measure for  $\mu$ -almost every  $x$ . By (2.8) and Proposition 4.2, for  $\mu$ -almost every  $x$  we have

$$\int v^b \otimes v^b d\sigma_x([v]) = \tau^b \otimes \tau^b, \quad \int (1 - \langle v, \tau \rangle^2) d\sigma_x([v]) = 0.$$

Thus  $\sigma_x = \delta_{[\tau(x)]}$ , identifying every subsequential limit.

Since  $0 \leq (\alpha_{\varepsilon})_+ \leq Ce_{\varepsilon}$ , any local weak limit of  $(\alpha_{\varepsilon}(x))_+ d\text{vol}_g(x) \delta_{[\nu(x)]}$  is dominated by  $C$  times the preceding limit. Its base measure is  $\mu$  by Proposition 4.2, so it equals  $\mu(dx)\delta_{[\tau(x)]}$ . Finally, oddness makes  $\Xi(x, v)(v)$  a continuous function on  $\mathbb{P}(T\Omega)$ , and

$$\int_{\Omega} \Xi(x, \nu)(q_{\varepsilon}) d\text{vol}_g - \int_{\Omega} (\alpha_{\varepsilon})_+ \Xi(x, \nu)(\nu) d\text{vol}_g \longrightarrow 0$$

by the vanishing negative and transverse parts on the compact base support of  $\Xi$ . Apply the convergence just proved to the second integral.

For  $\varphi \in C_c^{\infty}(S\Omega)$ , let  $d_x^H \varphi$  and  $d_v \varphi$  denote its horizontal and vertical derivatives. Where  $\nu$  is smooth, the chain rule gives

$$d(\varphi(x, \nu(x)))(q_{\varepsilon}) = d_x^H \varphi(x, \nu)(q_{\varepsilon}) + d_v \varphi(x, \nu)(\nabla_{q_{\varepsilon}} \nu).$$

For sufficiently small  $\varepsilon$ ,  $\nu$  is smooth near the intersection of  $\text{spt } q_\varepsilon$  with the compact base support of  $\varphi$ . Multiply  $\varphi(x, \nu(x))$  by a smooth cutoff supported where  $\nu$  is smooth and equal to one near this intersection, then extend by zero. The derivative of the cutoff has zero pairing with  $q_\varepsilon$  wherever  $\varphi$  can be nonzero. Since  $\text{div } q_\varepsilon = 0$ ,

$$\int_{\Omega} d_x^H \varphi(x, \nu)(q_\varepsilon) d\text{vol}_g = - \int_{\Omega} d_v \varphi(x, \nu)(\nabla_{q_\varepsilon} \nu) d\text{vol}_g. \quad (4.5)$$

We claim that, at every vertex  $x_0 \in \Omega$  of the limiting network, the outgoing directions  $v_i$  and positive integer multiplicities  $m_i$  satisfy

$$\sum_i m_i \delta_{v_i} = \sum_i m_i \delta_{-v_i}. \quad (4.6)$$

Choose  $r > 0$  small enough that  $\overline{B_r(x_0)} \subset \Omega$  and the network in this normal ball consists exactly of the finitely many incident geodesic arms. Given a smooth odd function  $a: S_{x_0}M \rightarrow \mathbf{R}$ , choose a smooth cutoff  $\zeta$  equal to one near zero and zero near  $[1, \infty)$ , and set

$$\varphi_r(x, v) := \zeta(\text{dist}(x, x_0)/r) a(P_{x \rightarrow x_0} v),$$

where  $P_{x \rightarrow x_0}$  is radial parallel transport. Extend by zero outside the ball. Then  $\varphi_r \in C_c^\infty(S\Omega)$  is odd in  $v$ , and  $\|d_v \varphi_r\|_\infty \leq C_a$  independently of  $r$ .

For fixed  $r$ , apply the preceding calculations to  $\Xi = d_x^H \varphi_r$ . By (4.5) and (4.3),

$$\left| \int_{\Omega} d_x^H \varphi_r(x, \tau)(\tau) d\mu \right| \leq C_a \sqrt{r}. \quad (4.7)$$

The integrand is independent of the sign of  $\tau$ . On the  $i$ th outward arm  $\gamma_i(t) = \exp_{x_0}(tv_i)$ , radial parallel transport makes  $\dot{\gamma}_i$  constant, so

$$d_x^H \varphi_r(\gamma_i(t), \dot{\gamma}_i(t))(\dot{\gamma}_i(t)) = r^{-1} \zeta'(t/r) a(v_i).$$

Its integral over the arm is  $-a(v_i)$ . Since  $\mu = 2\pi\|V\|$ , (4.7) gives

$$2\pi \left| \sum_i m_i a(v_i) \right| \leq C_a \sqrt{r}.$$

Let  $r \downarrow 0$ . For any smooth  $b$ , choose  $a(v) = b(v) - b(-v)$  to obtain (4.6). Pairing opposite arms with matching multiplicities and uniqueness for the geodesic equation give smooth continuation through every vertex, with no interior endpoints. When

$\Omega = M$  is closed, local finiteness and compactness give a finite network. Resolving its vertices by these pairings decomposes the network, with multiplicity, into finitely many closed immersed geodesics. This proves the theorem.  $\square$

**4.1. The integral curvature estimates.** We now prove Propositions 4.1 and 4.2 for the sequence fixed above.

Recall that  $\nu = (*F_\nabla)^\sharp/|F_\nabla|$  on  $\{|F_\nabla| > 0\}$ . One of the main tools in the analysis is the following defect:

$$\mathcal{S}_\varepsilon := |\nabla u|^2 - (*K)(\nu) + \varepsilon^2 |F_\nabla| |\nabla \nu|^2$$

on this set, and extend it by  $|\nabla u|^2$  where  $F_\nabla = 0$ . In an oriented orthonormal frame with  $e_3 = \nu$ , completing the square gives

$$|\nabla u|^2 - (*K)(\nu) = |\nabla_{e_3} u|^2 + |\nabla_{e_1} u + i \nabla_{e_2} u|^2 \geq 0. \quad (4.8)$$

In particular,  $\mathcal{S}_\varepsilon \geq 0$ . It vanishes in a neighbourhood if the Bogomolny equations hold in the curvature direction  $\nu$  and  $\nu$  is parallel. For critical points, the defect appears in the equation for the discrepancy: on  $\{|F_\nabla| > 0\}$ ,

$$(\Delta - \varepsilon^{-2}|u|^2)\xi_\varepsilon = \varepsilon^{-1}\mathcal{S}_\varepsilon + \varepsilon|F_\nabla| \operatorname{Ric}(\nu, \nu). \quad (4.9)$$

Using the bounds already obtained for the discrepancy for stable critical pairs and (4.9), we prove the following proposition which will be the key estimate in proving the localised integral curvature estimates of Proposition 4.1.

**Proposition 4.4** (Defect estimates). *Let  $(u, \nabla)$  be a smooth critical point in  $\Omega$ , stable under compactly supported variations, with  $|u| \leq 1$  and  $E_\varepsilon(u, \nabla; \Omega) \leq \Lambda$ . For every  $\Omega' \Subset \Omega$  and all sufficiently small  $\varepsilon$ ,*

$$\varepsilon^{-1} \int_{\Omega} \eta h \mathcal{S}_\varepsilon \, d\operatorname{vol}_g \leq C E_\varepsilon(u, \nabla; \Omega_0)^{1/2} (1 + \varepsilon \|d\eta\|_{L^\infty}) + C \varepsilon E_\varepsilon(u, \nabla; \Omega_0), \quad (4.10)$$

for  $0 \leq \eta \leq 1$  smooth with compact support in an open set  $\Omega_0 \subset \Omega'$ . The constant depends on the fixed interior domains, their geometry and  $\Lambda$ , but not on  $\Omega_0$ ,  $\eta$  or  $\varepsilon$ . Since  $h \geq c_0$  on  $\{|u|^2 \leq b_0\}$ , the same right-hand side, with a different constant, bounds

$$\varepsilon^{-1} \int_{\Omega \cap \{|u|^2 \leq b_0\}} \eta \mathcal{S}_\varepsilon \, d\operatorname{vol}_g.$$

Moreover,

$$\int_{\Omega' \cap \{|F_\nabla| > 0\}} \varepsilon^2 |F_\nabla|^2 |\nabla_\nu \nu|^2 d\text{vol}_g \leq C. \quad (4.11)$$

Along the sequence defining  $\mu$ , we also have

$$\int_{\Omega'} \mathcal{S}_\varepsilon d\text{vol}_g \longrightarrow 0, \quad (4.12)$$

and, for any measurable unit extension of  $\nu$  to  $\{F_\nabla = 0\}$ ,

$$\int_{\Omega'} (|\nabla u|^2 - (*K)(\nu)) d\text{vol}_g \longrightarrow 0. \quad (4.13)$$

Before proving this proposition, we show how its estimates imply Propositions 4.1 and 4.2. First we identify the limiting mass of the modified Jacobian and its associated vector field  $q_\varepsilon$ , and then the limit of the stress-energy tensor.

*Proof of Proposition 4.2.* First note that:

$$\begin{aligned} \alpha_\varepsilon &= (1 - t(|u|^2))|F_\nabla| + t'(|u|^2)(*K)(\nu), \\ R_\varepsilon &= t'(|u|^2)((*K)^\sharp - (*K)(\nu)\nu). \end{aligned} \quad (4.14)$$

Where the coefficients in (4.1) are nonzero,  $h \geq c_0$  and  $2h|F_\nabla| \leq e_\varepsilon$ . Together with  $|K| \leq |\nabla u|^2$ , this proves that  $|q_\varepsilon| \leq C e_\varepsilon$ . Since  $1 - t \geq 0$  and  $t' \geq 0$ ,

$$(\alpha_\varepsilon)_- \leq C(|\nabla u|^2 - (*K)(\nu)) \leq C\mathcal{S}_\varepsilon.$$

The integral of the right-hand side over every interior domain tends to zero by (4.12). In an oriented orthonormal frame with  $e_3 = \nu$ , the components of  $(*K)^\sharp$  perpendicular to  $\nu$  contain  $\nabla_{e_3} u$ . Hence

$$|R_\varepsilon| \leq C|\nabla u||\nabla_\nu u| \leq C|\nabla u|(|\nabla u|^2 - (*K)(\nu))^{1/2}. \quad (4.15)$$

Integrating on  $\Omega'$  and applying (4.13), the local energy bound and Cauchy–Schwarz, we obtain

$$\int_{\Omega'} ((\alpha_\varepsilon)_- + |R_\varepsilon|) d\text{vol}_g \longrightarrow 0.$$

Since  $t(a) = a\chi(a)$ , the scalar equation and the chain rule give

$$\begin{aligned} \frac{1}{2}\Delta \left( \int_1^{|u|^2} \chi(a) da \right) &= t'(|u|^2)|\nabla u|^2 - \varepsilon^{-2}t(|u|^2)h \\ &\quad + \frac{1}{2}\chi'(|u|^2)(|d|u|^2|^2 - 2|u|^2|\nabla u|^2). \end{aligned}$$

Substitution in (4.14) gives

$$\begin{aligned} \alpha_\varepsilon &= \varepsilon^{-2}h + \frac{1}{2}\Delta \left( \int_1^{|u|^2} \chi(a) da \right) + (1 - t(|u|^2))\varepsilon^{-1}\xi_\varepsilon \\ &\quad - t'(|u|^2)(|\nabla u|^2 - (*K)(\nu)) \\ &\quad - \frac{1}{2}\chi'(|u|^2)(|d|u|^2|^2 - 2|u|^2|\nabla u|^2). \end{aligned} \tag{4.16}$$

The last term is controlled by the first-order defect:

$$|d|u|^2|^2 - 2|u|^2|\nabla u|^2 \leq C|u|^2|\nabla u|(|\nabla u|^2 - (*K)(\nu))^{1/2}.$$

Indeed, in a local unitary trivialisation and an oriented orthonormal frame with  $e_3 = \nu$ ,

$$\begin{aligned} |d|u|^2|^2 - 2|u|^2|\nabla^A u|^2 &= 2 \left\langle \sum_{i=1}^3 (\nabla_{e_i}^A u)^2, u^2 \right\rangle, \\ \sum_{i=1}^3 (\nabla_{e_i}^A u)^2 &= (\nabla_{e_1}^A u + i\nabla_{e_2}^A u)(\nabla_{e_1}^A u - i\nabla_{e_2}^A u) + (\nabla_{e_3}^A u)^2. \end{aligned}$$

The first-order defect is given by (4.8). Since  $a\chi'(a)$  is bounded, (3.3), (4.13) and the energy bound show that the last three terms in (4.16) tend to zero in  $L_{\text{loc}}^1(\Omega)$ .

Moreover,  $|u|^2 \rightarrow 1$  in  $L_{\text{loc}}^1(\Omega)$  and  $\chi$  is bounded, so  $\int_1^{|u|^2} \chi(a) da \rightarrow 0$  in  $L_{\text{loc}}^1(\Omega)$ . Its Laplacian tends to zero against compactly supported smooth tests. By (3.4),  $\alpha_\varepsilon d\text{vol}_g \rightarrow \mu$  distributionally. The local total variation bound  $|q_\varepsilon| \leq Ce_\varepsilon$  and the vanishing negative part give convergence of the positive parts as Radon measures. Finally,

$$||q_\varepsilon| - (\alpha_\varepsilon)_+| \leq |R_\varepsilon| + (\alpha_\varepsilon)_-,$$

which proves the second convergence.

Let  $P = g - \nu^\flat \otimes \nu^\flat$ . The planar algebra gives

$$|2S - |\nabla u|^2 P| \leq C|\nabla u|(|\nabla u|^2 - (*K)(\nu))^{1/2}.$$

To see this, work in a local unitary trivialisation and an oriented orthonormal frame with  $e_3 = \nu$ . Project  $(\nabla_{e_1}^A u, \nabla_{e_2}^A u, \nabla_{e_3}^A u)$  onto the complex line  $\{(z, iz, 0) : z \in \mathbf{C}\}$ . The squared distance to this line is bounded by the defect in (4.8), and on the line  $2S = |\nabla^A u|^2 P$ . The change in the quadratic tensor is bounded by  $C|\nabla^A u|$  times that distance.

Since  $F^\top F = |F_\nabla|^2 P$ , the definition of  $T_\varepsilon$  gives

$$T_\varepsilon - e_\varepsilon \nu^\flat \otimes \nu^\flat = |\nabla u|^2 P - 2S + (\varepsilon^{-2} h^2 - \varepsilon^2 |F_\nabla|^2) P.$$

Consequently,

$$|T_\varepsilon - e_\varepsilon \nu^\flat \otimes \nu^\flat| \leq C|\nabla u|(|\nabla u|^2 - (*K)(\nu))^{1/2} + C\varepsilon^{-1}|\xi_\varepsilon|(h + \varepsilon^2 |F_\nabla|).$$

The first term tends to zero in  $L^1(\Omega')$  by (4.13) and the energy bound. For the second term, (3.2) and Cauchy–Schwarz give an  $L^1(\Omega')$  bound  $C\varepsilon$ , since  $\|h + \varepsilon^2 |F_\nabla|\|_{L^2} \leq C\varepsilon$ . The same algebra applies on the zero set with the chosen extension of  $\nu$ .  $\square$

*Proof of Proposition 4.1.* By (4.4),

$$\nabla_{q_\varepsilon} \nu = \alpha_\varepsilon \nabla_\nu \nu + \nabla_{R_\varepsilon} \nu.$$

On the core,  $\varepsilon^2 |F_\nabla| \geq c > 0$  and  $e_\varepsilon \leq C\varepsilon^{-2}$ . Since  $|\alpha_\varepsilon| \leq |q_\varepsilon| \leq Ce_\varepsilon$ , we have  $\varepsilon^{-2} \alpha_\varepsilon^2 / |F_\nabla|^2 \leq Ce_\varepsilon$ . Thus (4.11) and Cauchy–Schwarz give

$$\begin{aligned} \int_{B_r(x)} |\alpha_\varepsilon \nabla_\nu \nu| d\text{vol}_g &\leq \left( \int_{\Omega' \cap \{|F_\nabla| > 0\}} \varepsilon^2 |F_\nabla|^2 |\nabla_\nu \nu|^2 d\text{vol}_g \right)^{1/2} \\ &\cdot \left( \int_{B_r(x)} \varepsilon^{-2} \frac{\alpha_\varepsilon^2}{|F_\nabla|^2} d\text{vol}_g \right)^{1/2} \leq CE_\varepsilon(u, \nabla; B_r(x))^{1/2}. \end{aligned} \tag{4.17}$$

The ratios are extended by zero off the core.

For the transverse contribution, (4.15) gives on the core

$$\begin{aligned} |\nabla_{R_\varepsilon} \nu| &\leq C|\nabla u| |\nabla_\nu u| |\nabla \nu| \\ &\leq C\varepsilon^{-1} (|\nabla_\nu u|^2 + \varepsilon^2 |F_\nabla| |\nabla \nu|^2) \leq C\varepsilon^{-1} h \mathcal{S}_\varepsilon. \end{aligned}$$

Here  $h \geq c_0$  on the core. Choose a cutoff equal to one on  $B_r(x)$ , supported in  $B_{2r}(x)$ , with  $|d\eta| \leq C/r$ , and apply (4.10). Combining it with (4.17) proves (4.2). The local energy growth bound and the limit  $\varepsilon \rightarrow 0$  at fixed  $r$  give (4.3).  $\square$

The only task left is to prove the main estimate.

*Proof of Proposition 4.4.* First we show that in every  $\Omega' \Subset \Omega$ ,

$$\|\xi_\varepsilon\|_{L^\infty(\Omega')} \leq C\varepsilon^{-4/5}. \quad (4.18)$$

In particular, for each fixed  $c > 0$  and all sufficiently small  $\varepsilon$ ,

$$\varepsilon^2 |F_\nabla| \geq c/2 \quad \text{on } \Omega' \cap \{h \geq c\}. \quad (4.19)$$

The smallness of  $\varepsilon\|\xi_\varepsilon\|_{L^\infty}$  obtained here uses dimension three. Indeed, assuming analogous  $L^2$  and Lipschitz bounds in dimension  $n$ , the interpolation argument below would give

$$\varepsilon\|\xi_\varepsilon\|_{L^\infty} \leq C\varepsilon^{(4-n)/(n+2)},$$

which yields smallness for  $n < 4$ .

Choose  $\Omega' \Subset \Omega'' \Subset \Omega$ . Equations (2.5) and (2.6) give  $\text{Lip}(\xi_\varepsilon) \leq C\varepsilon^{-2}$  on  $\Omega''$ , and  $\|\xi_\varepsilon\|_{L^\infty(\Omega'')} \leq C\varepsilon^{-1}$ . If  $\|\xi_\varepsilon\|_{L^\infty(\Omega')} > 0$ , a ball centred in  $\overline{\Omega'}$ , of radius comparable to  $\varepsilon^2\|\xi_\varepsilon\|_{L^\infty(\Omega')}$ , lies in  $\Omega''$  and satisfies  $|\xi_\varepsilon| \geq \|\xi_\varepsilon\|_{L^\infty(\Omega')}/2$ . Consequently,

$$c\varepsilon^6\|\xi_\varepsilon\|_{L^\infty(\Omega')}^5 \leq \int_{\Omega''} \xi_\varepsilon^2 d\text{vol}_g \leq C\varepsilon^2.$$

This proves (4.18); (4.19) follows from  $\varepsilon^2 |F_\nabla| = h + \varepsilon\xi_\varepsilon$  and  $\varepsilon\|\xi_\varepsilon\|_{L^\infty(\Omega')} \rightarrow 0$ . Next we prove (4.11). Choose  $\zeta \in C_c^\infty(\Omega)$  equal to one on a neighbourhood of  $\overline{\Omega'}$ . For the symmetric bilinear form  $B$ , set

$$(\text{div } B)(Y) = \sum_i (\nabla_{e_i} B)(e_i, Y).$$

The gauge expression for  $\mathbf{W}_a$  is

$$G[\mathbf{W}_a] = (\text{div } B - \varepsilon^{-2}dh)(X_a) + \sum_i B(e_i, \nabla_{e_i} X_a).$$



By (2.11), the mixed terms vanish after summing the squares in  $a$ , so  $\sum_a G[\mathbf{W}_a]^2 \geq |\operatorname{div} B - \varepsilon^{-2}dh|^2$ . Moreover,

$$G[\zeta \mathbf{W}_a] = \zeta G[\mathbf{W}_a] + B(X_a, \nabla \zeta).$$

Apply stability and (2.10) to the compactly supported pairs  $\zeta \mathbf{W}_a$ , and use (2.15). Since  $\varepsilon^2|B|^2 \leq Ce_\varepsilon$ , the cutoff term is bounded by the energy on  $\operatorname{spt} \zeta$ . It follows that

$$\int_{\Omega'} \varepsilon^2 |\operatorname{div} B - \varepsilon^{-2}dh|^2 d\operatorname{vol}_g \leq C.$$

Since  $dF_\nabla = 0$ , the vector field  $|F_\nabla|\nu$  is divergence-free. On  $\{|F_\nabla| > 0\}$ ,

$$\operatorname{div} B = d|F_\nabla| - |F_\nabla|(\nabla_\nu \nu)^\flat, \quad \operatorname{div} B - \varepsilon^{-2}dh = -|F_\nabla|(\nabla_\nu \nu)^\flat + \varepsilon^{-1}d\xi_\varepsilon.$$

Now use the interior bound for  $d\xi_\varepsilon$  from Theorem 3.1 and conclude with (4.11). It remains to prove (4.10), using (4.9). Indeed for every nonnegative  $\phi \in C_c^\infty(\Omega)$ ,

$$\int_\Omega \phi \mathcal{S}_\varepsilon d\operatorname{vol}_g \leq -\varepsilon \int_\Omega (\langle d\xi_\varepsilon, d\phi \rangle + \varepsilon^{-2}|u|^2 \xi_\varepsilon \phi) d\operatorname{vol}_g + C\varepsilon^2 \int_\Omega \phi |F_\nabla| d\operatorname{vol}_g, \quad (4.20)$$

where  $C$  bounds the curvature on  $\operatorname{spt} \phi$ . To justify this inequality across  $\{F_\nabla = 0\}$ , write  $b = (*F_\nabla)^\sharp$  and regularise its norm by

$$f_\delta = \sqrt{|b|^2 + \delta^2}, \quad \xi_{\varepsilon,\delta} = \varepsilon f_\delta - \varepsilon^{-1}h.$$

The curvature equation and the one-form Weitzenböck formula give the regularised defect

$$\mathcal{S}_{\varepsilon,\delta} = |\nabla u|^2 - \frac{\langle (*K)^\sharp, b \rangle}{f_\delta} + \frac{\varepsilon^2}{f_\delta} (|\nabla b|^2 - |df_\delta|^2) \geq 0.$$

The inequality follows from  $|K| \leq |\nabla u|^2$  and  $|df_\delta| \leq |\nabla b|$ . The regularised version of (4.20) has an additional error bounded by  $\delta \int_\Omega \phi d\operatorname{vol}_g$ . As  $\delta \rightarrow 0$ ,  $\xi_{\varepsilon,\delta} \rightarrow \xi_\varepsilon$  in  $H_{\operatorname{loc}}^1$  for fixed  $\varepsilon$ . Off the zero set,  $\mathcal{S}_{\varepsilon,\delta} \rightarrow \mathcal{S}_\varepsilon$ , while on the zero set its lower limit is at least  $|\nabla u|^2$ . Fatou's lemma therefore proves (4.20).

Choose a fixed nonnegative  $\phi \in C_c^\infty(\Omega)$  equal to one near  $\overline{\Omega'}$ . Integration by parts gives

$$\begin{aligned} \int_{\Omega'} \mathcal{S}_\varepsilon d\text{vol}_g &\leq \varepsilon \int_\Omega \xi_\varepsilon \Delta \phi d\text{vol}_g - \varepsilon^{-1} \int_\Omega |u|^2 \xi_\varepsilon \phi d\text{vol}_g \\ &\quad + C\varepsilon^2 \int_\Omega \phi |F_\nabla| d\text{vol}_g \longrightarrow 0. \end{aligned}$$

Here (3.3) controls the first two terms and (3.4) controls the last. This proves (4.12). The first-order defect is bounded by  $\mathcal{S}_\varepsilon$  off the zero set and by  $2\mathcal{S}_\varepsilon$  on it, proving (4.13) for every measurable unit extension of  $\nu$ .

Finally, since  $h \geq 0$ , test (4.20) with  $\eta h$ , use the scalar equation and integrate by parts:

$$\begin{aligned} \int_\Omega \eta h \mathcal{S}_\varepsilon d\text{vol}_g &\leq -\varepsilon \int_\Omega \eta \xi_\varepsilon |\nabla u|^2 d\text{vol}_g + \varepsilon \int_\Omega \langle \xi_\varepsilon dh - h d\xi_\varepsilon, d\eta \rangle d\text{vol}_g \\ &\quad + C\varepsilon^2 \int_\Omega \eta h |F_\nabla| d\text{vol}_g. \end{aligned}$$

By the pointwise and discrepancy estimates,

$$\varepsilon \left| \int_\Omega \eta \xi_\varepsilon |\nabla u|^2 d\text{vol}_g \right| \leq \|\varepsilon^{-1} \xi_\varepsilon\|_{L^2(\Omega')} \left( \int_{\Omega_0} \varepsilon^4 |\nabla u|^4 d\text{vol}_g \right)^{1/2} \leq C\varepsilon E_\varepsilon(u, \nabla; \Omega_0)^{1/2}.$$

Also  $\|h\|_{L^2(\Omega_0)} \leq C\varepsilon E_\varepsilon(u, \nabla; \Omega_0)^{1/2}$ ,  $|dh| \leq |\nabla u|$ ,  $\|d\xi_\varepsilon\|_{L^2(\Omega')} \leq C$  and  $\|\xi_\varepsilon\|_{L^2(\Omega')} \leq C\varepsilon$ . The cutoff term is bounded by  $C\varepsilon^2 \|d\eta\|_{L^\infty} E_\varepsilon(u, \nabla; \Omega_0)^{1/2}$ . Finally,  $2h|F_\nabla| \leq e_\varepsilon$ , so the curvature term is at most  $C\varepsilon^2 E_\varepsilon(u, \nabla; \Omega_0)$ . Divide by  $\varepsilon$  to obtain (4.10).  $\square$

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